

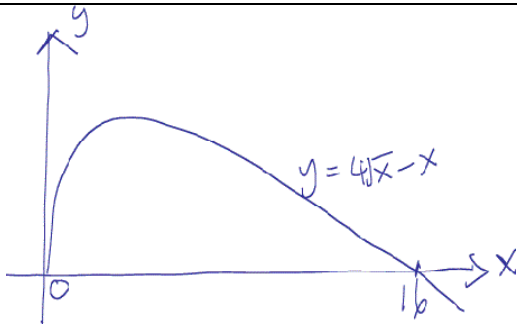
JC2 2013 H1 Mathematics Prelim Suggested Solutions

1	$\ln(e^{3x} - 3e^{2x}) = x + 2\ln 2$ $\ln(e^{3x} - 3e^{2x}) - \ln(2^2) = x$ $\ln\left(\frac{e^{3x} - 3e^{2x}}{4}\right) = x$ $\frac{e^{3x} - 3e^{2x}}{4} = e^x$ $e^{3x} - 3e^{2x} = 4e^x$ $e^{3x} - 3e^{2x} - 4e^x = 0$ $e^x(e^{2x} - 3e^x - 4) = 0$ $e^x = 0 \text{ (NA) or } e^{2x} - 3e^x - 4 = 0$ Let $y = e^x$ $y^2 - 3y - 4 = 0$ $(y+1)(y-4) = 0$ $y = -1 \text{ or } y = 4$ $e^x = -1 \text{ (NA) or } e^x = 4$ $x = \ln 4$
2	$kx^2 + x + k > 4 - 2x$ $kx^2 + 3x + k - 4 > 0$ (For quadratic curve to be positive, it must not cut the x-axis, i.e. no real solution, $b^2 - 4ac < 0$ and above x axis) $3^2 - 4(k)(k-4) < 0$ $k > 0 \text{ and } -4k^2 + 16k + 9 < 0$ $k > \frac{9}{2} \text{ or } k < -\frac{1}{2}$ thus, $k > \frac{9}{2}$
3	(a) $\int_0^2 e^{-3x} dx = \int_0^2 e^e (e^{-3x}) dx$

$$= e^e \left[\frac{e^{-3x}}{-3} \right]_0^2 = -\frac{1}{3} e^e [e^{-6} - 1]$$

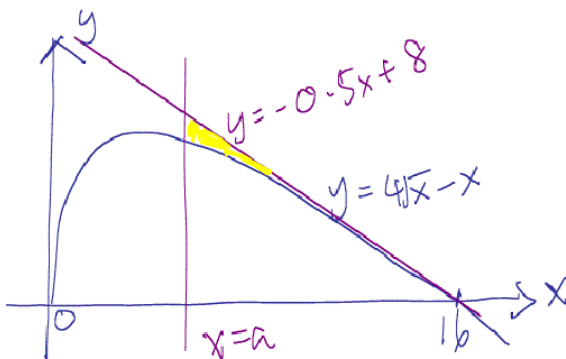
$$\begin{aligned} \text{(b)} \quad \int \left(\frac{a}{(3x+1)} + \frac{x^{5/2}}{\sqrt{x}} \right) dx &= \int \left(\frac{a}{(3x+1)} + x^2 \right) dx \\ &= \frac{a}{3} \ln |3x+1| + \frac{1}{3} x^3 + c \end{aligned}$$

4



Using GC, $\frac{d}{dx}(4\sqrt{x} - x) \Big|_{x=16} = -0.5$

Thus equation of tangent is $y = -0.5x + 8$



Area bounded = 1.673

$$\int_a^{16} -0.5x + 8 - (4\sqrt{x} - x) dx = 1.673 \text{ units}^2$$

Using GC, $a = 8$

5

(i) $y = e^{2x} - 7x$

$$\frac{dy}{dx} = 2e^{2x} - 7$$

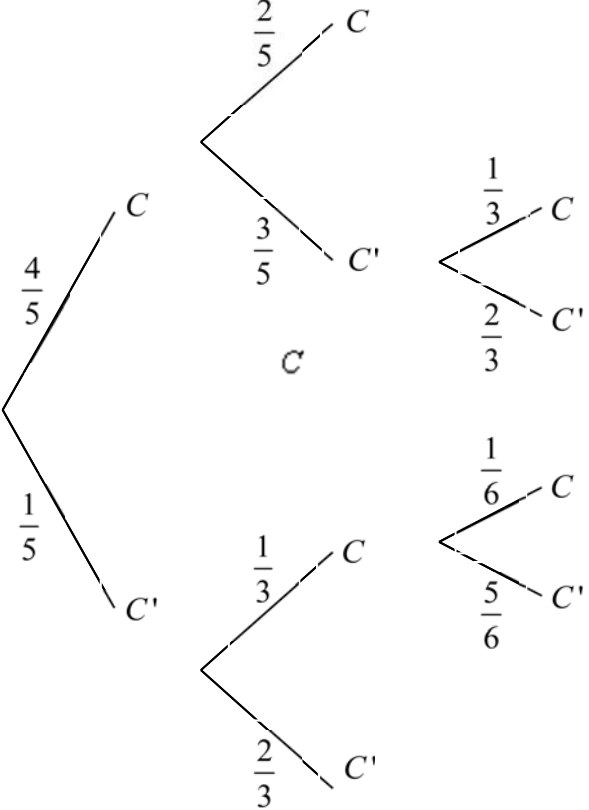
So $-\frac{1}{2e^{2x} - 7} = -1$

$$e^{2x} = 4$$

$$2x = \ln 4 = \ln 2^2 = 2 \ln 2$$

	$x = \ln 2$ $y = e^{\ln 4} - 7 \ln 2 = 4 - 7 \ln 2$ So point C is $(\ln 2, 4 - 7 \ln 2)$ (ii) Equation of Normal: $-1 = \frac{y - (4 - 7 \ln 2)}{x - (\ln 2)}$ $-x + \ln 2 = y - 4 + 7 \ln 2$ $y = -x + 4 - 6 \ln 2$ So point A is $(0, 4 - 6 \ln 2)$ $\text{Area of triangle OAC} = \frac{1}{2} (4 - 6 \ln 2) \ln 2$ $= 0.0551 \text{ (to 3 s.f.)}$				
6 (i)	By similar triangles, we have $\frac{r}{12} = \frac{18-h}{18}$, i.e., $h = \frac{3(12-r)}{2}$.				
(ii)	$V = \text{base area} \times \text{height}$ $= \pi r^2 \frac{3(12-r)}{2}$ $= \frac{3\pi r^2(12-r)}{2}.$				
(iii)	Differentiating $V = \frac{3\pi r^2(12-r)}{2} = \frac{3\pi}{2}(12r^2 - r^3)$ w.r.t. r , we have $\frac{dV}{dr} = \frac{3\pi}{2} [24r - 3r^2]$ $\frac{dV}{dr} = \frac{9\pi r}{2} (8-r).$ When $\frac{dV}{dr} = 0$, $r(8-r) = 0$, i.e., $r = 0$ (N.A.) or $r = 8$. 1st derivative test <table><tr><td>r</td><td>8⁻</td><td>8</td><td>8⁺</td></tr></table>	r	8 ⁻	8	8 ⁺
r	8 ⁻	8	8 ⁺		

	<table><tr><td>$\frac{dV}{dr}$</td><td>+ve</td><td>0</td><td>-ve</td></tr><tr><td>Slope of tgt</td><td></td><td></td><td></td></tr></table>	$\frac{dV}{dr}$	+ve	0	-ve	Slope of tgt			
$\frac{dV}{dr}$	+ve	0	-ve						
Slope of tgt									
	V is a maximum when $r = 8$. When $r = 8$, $V = \frac{3\pi(8)^2(12-8)}{2} = 384\pi$.								
7	(a) Not satisfactory because students who are given as an odd number will not have any chance of getting selected. Thus, process is not random. (b) Step 1: To obtain a quota sampling of 100 people, divide the people into two mutually exclusive subgroups: male and female. e.g. <table><tr><td>Male</td><td>Female</td></tr><tr><td>40</td><td>60</td></tr></table> Step 2: Surveyors can station themselves along the exits of MRT stations at peak hours. Step 3: Surveyors can then be given a free choice to interview the commuters to fill the quota. Advantage: Information can be collected quickly and conveniently.	Male	Female	40	60				
Male	Female								
40	60								
8	$P(B) = P(2,1,2) + P(1,2,2) + P(2,2,1) + P(3,1,1) + P(1,3,1) + P(1,1,3)$ $= 3\left(\frac{1}{6}\right)^3 + 3\left(\frac{1}{6}\right)^3 = \frac{1}{36}$ $P(B A) = \frac{P(B \cap A)}{P(A)} = \frac{2\left(\frac{1}{6}\right)^3}{\frac{1}{6}} = \frac{1}{18}$ Since $P(B) \neq P(B A)$, A and B are not independent.								

9(i))	
(ii)	$P(\text{clear quest successfully})$ $= P(\text{clear 2 stages})$ $= \frac{4}{5} \cdot \frac{2}{5} + \frac{4}{5} \cdot \frac{3}{5} \cdot \frac{1}{3} + \frac{1}{5} \cdot \frac{1}{3} \cdot \frac{1}{6}$ $= \frac{221}{450} \text{ or } 0.491(3sf).$
	$P(\text{at least 2 of 3 players clear the quest})$ $= P(\text{exactly 2 clear the quest}) + P(\text{all 3 clear the quest})$ $= 3 \times \left(\frac{221}{450}\right)^2 \left(1 - \frac{221}{450}\right) + \left(\frac{221}{450}\right)^3$ $= 0.487$
10	(a) Let X denote the number of defective projectors out of 20 projectors i.e. $X \sim B(20, 0.08)$ $P(X \leq 2) = 0.78795 = 0.788 \quad (3 \text{ s.f.})$

$$\begin{aligned}
 \text{(b)} \quad P(X \leq 6 | X > 3) &= \frac{P(3 < X \leq 6)}{P(X > 3)} = \frac{P(4 \leq X \leq 6)}{1 - P(X \leq 3)} \\
 &= \frac{P(X \leq 6) - P(X \leq 3)}{1 - P(X \leq 3)} \\
 &= \frac{0.99936 - 0.92938}{0.070615} = 0.99097 \\
 &= 0.991 \quad (3 \text{ s.f.})
 \end{aligned}$$

$$\text{(c)} \quad E(X) = 20(0.08) = 1.6, \quad \text{Var}(X) = 20(0.08)(0.92) = 1.472$$

Let $\bar{X}$ be the mean number of defective projectors in 65 batches.

Since $n = 65$ is large, by the Central Limit Theorem,

$$\bar{X} \sim N\left(1.6, \frac{1.472}{65}\right), \quad \text{where } \bar{X} = \frac{X_1 + X_2 + \dots + X_{65}}{65}$$

$$P(\bar{X} \leq 2) = 0.99607 = 0.996 \quad (3 \text{ s.f.})$$

Y : no of defective projectors out of 120 projectors.

$$Y \sim B(120, 0.08)$$

Since $n = 120$ is large, $np = 9.6 > 5$, $nq = 110.4 > 5$

$Y \sim N(9.6, 8.832)$ approx

$$P(Y \geq a) > 0.6$$

$$\xrightarrow{\text{c.c.}} P(Y > a - 0.5) > 0.6$$

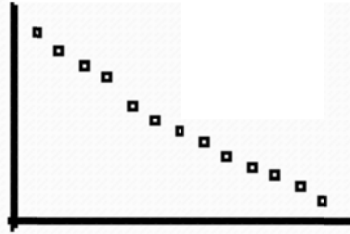
a	$P(Y > a - 0.5)$
9	0.644
10	0.513

Largest $a = 9$

11	$X \sim N(\mu, \frac{3\mu}{7})$ $P(X > 2\mu) = 0.1016$ $P(Z > \frac{2\mu - \mu}{\sigma}) = 0.1016$ $P(Z > \frac{\mu}{\sqrt{\frac{3}{7}}\mu}) = 0.1016$ $P(Z > \sqrt{\frac{7\mu}{3}}) = 0.1016$ $P(Z < \sqrt{\frac{7\mu}{3}}) = 0.8984$ $\sqrt{\frac{7\mu}{3}} = 1.2725$ $\mu = 0.694, \sigma = 0.545$
12	Let B and C be the mass of a box and a can of drink respectively. $B \sim N(200, 8^2) \text{ and } C \sim N(510, 14^2)$ (i) $P(2200 < C_1 + C_2 + C_3 + C_4 + B < 2300)$? $C_1 + C_2 + C_3 + C_4 + B \sim N(4(510) + 200, 4(14^2) + 8^2)$ $\Rightarrow C_1 + C_2 + C_3 + C_4 + B \sim N(2240, 848)$ $P(2200 < C_1 + C_2 + C_3 + C_4 + B < 2300) = 0.896$ (ii) The mass of a can and the mass of a box are independent of each other. (iii) $P(C_1 - C_2 > 20) = ?$ $C_1 - C_2 \sim N(510 - 510, 14^2 + 14^2)$ $C_1 - C_2 \sim N(0, 392)$ $P(C_1 - C_2 > 20) = 2P(C_1 - C_2 > 20) = 0.312$
13	(a) x, moisture (content) is controlled in this experiment. (b) Least square regression line of y on x: $y = 5.3538 - 0.075824x$ $y = 5.35 - 0.0758x \quad (3 \text{ s.f.})$

Links

$y = a + bx$
 $a = 5.353846154$
 $b = -.0758241758$
 $r^2 = .9849149345$
 $r = -.9924288057$



(c) a : calorific value of wood with zero/no moisture or dry maximum calorific value.

b : each 1(%) rise in moisture content reduces calorific value by 0.0758 MWh/tonne.

(d) $y = 5.3538 - 0.075824(27)$
 $= 3.3065 = 3.31$ (3 s.f.)

(e) $y = 5.3538 - 0.075824(80)$
 $= -0.71212 = -0.712$ (3 s.f.)

Negative value for calorific value is impossible or more energy needed than is generated.

14

(i) $\bar{x} = \frac{\sum x}{n} = \frac{26080}{80} = 326$

$$s^2 = \frac{1}{n-1} \left[\sum x^2 - \frac{(\sum x)^2}{n} \right] = \frac{1}{79} \left[8530200 - \frac{(26080)^2}{80} \right] = 355.95 = 356$$

(ii) $H_0: \mu = 330$, where μ is the population mean amount of beverage in a can

$$H_1: \mu < 330$$

@5% significant level, reject H_0 if $p < 0.05$

Apply Z-test,

$$\mu_0 = 330, \sigma = \sqrt{355.95}, \bar{x} = 326, n = 80, ,$$

From GC, $p\text{-value} = 0.028959$

Since $p\text{-value} = 0.028959 < 0.05$

We reject H_0 . Conclude that there is sufficient evidence at 5% significant level that there is evidence to suggest that the new machine has dispensed

lesser than 330ml

- (iii) No. Since n is large by CLT, the mean amount of beverage that the machine dispenses can be approximated to a normal distribution so no need to assume normal distribution.